

Design and simulation

As a device designer, you need test and measurement equipment right from the R&D stage to the deployment stage. Most of the development work for different sub-systems is done in parallel. Once the system is integrated, you need some other tests to check the workability of the entire system. Some of the common T&M instruments that are needed in the entire design flow are oscilloscopes, logic analysers, vector signal generators, spectrum analysers for swept measurements and vector signal analysers for modulation analysis.

Ideally, a base station design implementing WiMAX is expected to support air interfaces like WCDMA, CDMA2000 and GSM. It should be capable of handling single channel as well as multiple channels. The design of base station and simulation for above capabilities is critical and time-consuming.

Electronic design automation (EDA) tools used for designing a device now offer a mobile WiMAX library to help wireless systems designers and verification engineers to speed up the development of WiMAX devices. "The library provides signal processing models, preconfigured simulation set-up, signal sources and receivers to facilitate simulation of fully-coded systems," shares Sarpotdar. "Moreover, the library can provide transmitter measurements (such as error vector magnitude and complementary cumulative distribution function) and receiver measurements (such as receiver sensitivity and bit error rate) that directly help the designers," she adds.

Expressing his views on EDA tools, Mahesh Basavaraju, assistant manager, product support and applications, Rohde & Schwarz, says, "Using specialised EDA tools, engineers can see the interaction between architecture, circuit devices, parameters and RF impairments."

RF signal generation and verification

WiMAX operates in microwave radio frequencies. "The base and mobile sta-

tions have to be tested for their RF performance in order to maintain the best possible quality of radio link, using the least amount of carrier power," says Sarpotdar. For troubleshooting of the design and prototypes of transmitter and receiver, T&M equipment should be capable of working at the baseband as well as in the RF domain.

It is difficult to generate OFDM signals for test performance and even more difficult to decompose them. Many new vector signal generators directly help test engineers by generating the OFDM signals. However, you need vector signal analysers as you deal with multiple-carrier and OFDM signals, where accurate phase measurement is important. "Signal generators need to be flexible to create signals that designers wish to test their system with. These should also comply with the set standards so that designers can trust the test results," says Sarpotdar.

While dealing with multiple signals, it is important that phase distortions are corrected and the signal is recovered in time. The signal generators used must ensure fast set-up time and rapid switch-in frequencies and amplitudes.

A conventional signal generator is capable of generating one signal. You cannot be carrying multiple signal generators to test a MIMO system. "There equipment combine two independent vector signal generators in one box for 2x2 MIMO testing. You can simulate MIMO for conditions like fading and phase coherence that are very useful for various test cases," explains Basavaraju.

Signal decomposing and analysis

The complexity of implementing the MIMO capability in WiMAX presents unique measurement challenges. Test receivers should be capable of emulating multiple channels for MIMO test. Equipment are available with specialised and critical instrumentation for testing multiple antennae systems.

Signal analyser software today have a unique ability to acquire the

Parametric requirement for WiMAX T&M

1. Wide bandwidths
2. High-speed capture and analysis
3. Vector signal analysis (due to different modulation schemes)
4. Multiple-channel testing
5. OFDM testing
6. Repeatable and reliable results
7. Low phase noise

signal, analyse it and demodulate it before computing and displaying a wide range of useful parameters. Some equipment designed using state-of-the-art RF and digital signal processing (DSP) technologies are capable of performing such a task.

"The real-time spectrum analyser series, along with signal analysis software, provides in-depth measurement for troubleshooting of the WiMAX device at the design stage," says Antonio Policek, marketing manager, Asia Pacific, Tektronix. "Some high-performance signal analysers also go up to 20 GHz of frequencies, capturing and sampling frequencies up to 110 MHz," he adds.

According to Policek, there are some portable/handheld RF spectrum analysers that are used by regulatory bodies and administrative bodies to monitor the frequency usage.

There are a number of software tools that can generate OFDM waveforms. T&M manufacturers have made these tools easy to use by providing drop-down menus. You can configure the number of frames to be generated, zones, individual bursts, modulation format, etc.

Conformance tests and certification

The entire WiMAX deployment is a combined effort of WiMAX equipment manufacturers, WiMAX solution providers and operators providing WiMAX services. To avail the best quality of service and avoid issues like 'dropped call' and 'network busy,' all the devices must undergo tests and certifications as per the standards of WiMAX Forum.